# Mathieu Padlewski

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Active metamaterials · Topological physics · Wave-based computing

## Positions

**ESPCI Paris, PSL University** – Postdoc — Sep. 2026 – present
Institut Langevin
Project: *Time-Varying Media, Temporal Metamaterials and Wave Control*
Advisors: Prof. M. Fink, Prof. E. Fort

**École Polytechnique Fédérale de Lausanne (EPFL)** – Guest Scientist — Nov. 2025 – Aug. 2026
Laboratory of Wave Engineering (LWE)
Project: *Classical Emulation of Quantum Phenomena*

**EPFL** – Postdoc — Jul. 2025 – Oct. 2025
LWE (Collaboration with LG Electronics)
Project: *Active Noise Abatement of Electronic Appliances*
Advisors: Dr. S. Meyer, Dr. H. Lissek

**EPFL** – Research Intern — Mar. 2020 – Aug. 2021
Laboratory of Quantum Materials (QMAT)
Project: *Topological Semimetals under Torsional Strain*
Advisors: Dr. C. Putzke, Prof. P. Moll

## Education

**EPFL** – PhD in Physics — Sep. 2021 – Jun. 2025
Laboratory of Wave Engineering (LWE)
Thesis: *Active Acoustic Metamaterials: A New Way to Understand Nonlinear and Topological Phenomena*
Advisors: Dr. H. Lissek, Prof. R. Fleury

**EPFL** – MSc in Physics — Sep. 2017 – Dec. 2019
Physics Laboratory of Complex Matter (LPMC)
Master's project: *Charge Density Waves and Transport Properties of 4Hb-$TaS_2$*
Advisors: Dr. E. Martino, Prof. L. Forró

**University of Geneva** – BSc in Physics — Sep. 2013 – Jun. 2017

## Awards and Grants

**Seal of Excellence** – European Commission — Feb. 2026
Horizon Europe MSCA Postdoctoral Fellowship 2025, with CNRS
Project: *Wavebits: Analog Quantum-Inspired Computing*

## Publications

*Classical analog emulation of quantum circuits via time-averaged dynamic states* — Oct. 2026
M. Valle, T. Tuuva, B. Apffel, H. Lissek, R. Fleury, **M. Padlewski**, *Phys. Rev. Applied* (2026)

*Coexistence of broadband conduction and amplitude-driven topological confinement in synthetic acoustic matter* — Oct. 2026
**M. Padlewski**, M. Volery, R. Fleury, et al., *Phys. Rev. Applied* (2026)

*Observation of amplitude-driven nonreciprocity for energy guiding* — Mar. 2025
**M. Padlewski**, R. Fleury, H. Lissek, *Phys. Rev. B* **111**, 125156

*Practical realization of chiral nonlinearity for strong topological protection* Jan. 2025
X. Guo, L. Jezequel, **M. Padlewski**, et al., *SciPost Phys.* **18**, 034

*Active acoustic Su-Schrieffer-Heeger-like metamaterial* Jul. 2023
**M. Padlewski**, M. Volery, R. Fleury, et al., *Phys. Rev. Applied* **20**, 014022

*Evidence of a coupled electron-phonon liquid in $NbGe_2$* Sep. 2021
H.-Y. Yang, X. Yao, V. Plisson, et al., *Nat. Commun.* **12**, 5292

*Probing intraband excitations in $ZrTe_5$: A high-pressure infrared and transport study* Mar. 2020
D. Santos-Cottin, **M. Padlewski**, E. Martino, et al., *Phys. Rev. B* **101**, 125205

## Talks and Conferences

Institut Langevin, ESPCI Paris, France Sep. 2026
Seminar

IETR CNRS Institute, University of Rennes, France Jun. 2026
Invited talk

Wave Propagation and Control in Complex Media, Cargèse, France Apr. 2024
Selected conference talk

Forum Acusticum, Turin, Italy Sep. 2023
Conference talk

IRP Centre d'Acoustique Summer School, Sherbrooke, Canada Jun. 2023
Poster

Symposium on Acoustic Metamaterials, Nîmes, France Oct. 2022
Conference talk

## Outreach

Listen to quantum atoms talk together thanks to acoustics
Dimensions – EPFL Magazine

Scientists use sound waves to unlock the secrets of quantum physics
The Brighter Side of News

Listen to quantum acoustic atoms – a new way to study the secrets of the universe
IFLScience – LabX Media Group

Des ondes sonores pour explorer la mécanique quantique
La Revue Polytechnique – Polymedia SA

The science is sound
Shaastra – IIT Madras

## Peer Review

Physical Review Letters, Physical Review B, Communications Physics

## Programming

Python, Qiskit, MATLAB, Simulink, LabVIEW, Mathematica, COMSOL, Inventor, EAGLE

## Languages

French (native), English (native)

## Activities

Music (composition, guitar); endurance sports; useless Arduino projects